

IOWA FALLS, IA, USA

WMO: **722331**

Lat: **42.467N** Lon: **93.267W** Elev: **347** StdP: **97.23** Time Zone: **-6.00 (NAC)** Period: **07-19** WBAN: **54941**

Annual Heating, Humidification, and Ventilation Design Conditions

	Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF	
				99.6%			99%			0.4%		1%					
		99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(1)	1	-22.7	-20.1	-26.2	0.4	-22.5	-23.6	0.5	-19.9	15.3	-7.9	14.2	-9.0	3.9	290	0.570	(1)

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

	Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB		
			0.4%		1%		2%		0.4%		1%		2%				
			DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD	
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(2)	7	11.0	32.6	23.5	31.0	22.9	29.3	21.8	25.2	30.3	24.1	28.8	23.2	27.8	5.2	210	(2)

	Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB	
	0.4%			1%			2%			0.4%		1%		2%			
	DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(3)	23.7	19.3	28.3	22.7	18.1	27.1	21.8	17.2	26.3	78.8	30.4	74.4	29.0	70.9	27.8	27.8	(3)

Extreme Annual Design Conditions

	Extreme Annual WS				Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
					Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
	1%	2.5%	5%		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
	(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(4)	12.6	11.2	9.8	DB	-28.3	35.1	2.7	2.0	-30.3	36.6	-31.9	37.8	-33.4	38.9	-35.3	40.4	(4)
(5)				WB	-28.5	26.8	2.5	1.0	-30.3	27.5	-31.8	28.1	-33.2	28.6	-35.1	29.3	(5)

Monthly Climatic Design Conditions

				Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
				(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg		8.7	-7.8	-5.8	1.7	8.9	15.6	21.4	22.4	20.9	17.8	10.2	2.5	-4.4	(6)
(7)		DBStd		11.96	7.19	6.89	6.89	5.27	5.05	3.46	3.31	2.75	4.43	5.05	5.86	6.34	(7)
(8)		HDD10.0		2095	551	442	268	82	10	0	0	0	2	61	232	447	(8)
(9)		HDD18.3		3919	809	676	515	284	114	11	6	7	62	255	476	705	(9)
(10)		CDD10.0		1621	0	1	12	49	185	342	384	338	237	67	7	0	(10)
(11)		CDD18.3		402	0	0	1	1	31	102	131	87	47	3	0	0	(11)
(12)		CDH23.3		3654	0	0	7	34	307	933	1205	698	427	42	1	0	(12)
(13)		CDH26.7		1156	0	0	0	4	96	329	413	179	129	6	0	0	(13)
(14)	Wind	WSAvg		4.3	5.2	5.2	5.1	5.5	4.9	3.9	2.8	2.4	3.1	4.1	4.9	4.9	(14)
(15)	Precipitation	PrecAvg		859	24	25	52	84	106	128	112	104	85	60	45	29	(15)
(16)		PrecMax		1361	80	78	127	201	244	391	247	345	337	132	141	116	(16)
(17)		PrecMin		569	1	1	3	19	24	43	19	10	19	6	0	6	(17)
(18)		PrecStd		175	16	18	30	43	52	67	58	60	54	37	35	22	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB		8.9	14.7	24.0	27.0	32.9	34.1	34.1	32.6	32.5	27.3	21.0	12.7	(19)
(20)			MCWB		5.9	10.9	15.6	16.4	21.0	23.3	25.0	23.6	23.0	17.0	13.8	8.3	(20)
(21)		2%	DB		5.0	8.0	18.7	23.8	29.1	32.4	32.6	30.3	30.2	24.0	16.4	8.3	(21)
(22)			MCWB		3.0	5.7	13.0	14.8	19.0	22.8	24.6	23.3	21.8	15.9	11.7	6.3	(22)
(23)		5%	DB		2.8	5.3	15.0	20.9	26.9	30.2	30.7	28.7	27.8	21.4	13.1	6.0	(23)
(24)			MCWB		1.7	3.4	10.9	12.8	17.8	21.6	24.1	22.6	20.2	15.0	9.1	4.1	(24)
(25)		10%	DB		1.4	2.9	11.2	17.9	24.2	28.4	28.9	27.2	25.8	18.9	10.9	3.0	(25)
(26)			MCWB		0.6	1.7	8.0	11.7	17.1	20.4	22.6	21.1	19.0	13.9	7.6	1.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB		6.4	11.5	16.9	17.8	22.2	26.3	26.9	25.5	24.8	20.2	14.8	10.3	(27)
(28)			MCDB		9.3	15.5	22.6	24.3	29.2	32.1	31.9	30.8	30.6	23.2	18.4	11.8	(28)
(29)		2%	WB		3.2	6.5	14.1	15.9	20.8	24.4	25.5	24.1	23.1	18.2	12.2	6.6	(29)
(30)			MCDB		4.4	7.9	17.4	21.1	27.4	29.4	30.5	28.4	27.8	21.2	16.3	7.8	(30)
(31)		5%	WB		1.9	3.7	11.0	14.5	19.6	23.2	24.5	23.2	21.8	16.4	9.8	3.9	(31)
(32)			MCDB		2.9	5.3	14.5	19.7	24.6	28.1	29.3	27.3	26.1	19.8	12.4	5.9	(32)
(33)		10%	WB		0.7	1.7	8.1	12.7	18.4	22.0	23.5	22.3	20.6	14.4	7.6	2.0	(33)
(34)			MCDB		1.4	2.9	11.1	17.1	22.7	26.8	28.1	25.9	24.1	18.4	10.7	3.1	(34)
(35)	Mean Daily Temperature Range		MDBR		8.5	9.1	9.2	11.4	11.4	10.8	11.0	11.2	12.5	11.7	10.1	8.1	(35)
(36)		5% DB	MCDBR		9.0	10.4	15.1	16.6	14.3	13.0	12.1	12.1	13.9	15.2	14.3	10.4	(36)
(37)			MCWBR		7.7	8.3	9.7	8.9	6.3	5.7	6.0	6.2	6.6	8.2	9.4	8.1	(37)
(38)		5% WB	MCDBR		8.8	10.2	14.2	13.9	11.7	11.5	11.2	10.7	11.6	11.9	13.6	10.0	(38)
(39)			MCWBR		7.8	8.3	9.4	8.7	6.1	6.3	6.1	6.0	6.2	7.9	9.7	8.0	(39)
(40)	Clear-Sky Solar Irradiance	taub			0.278	0.295	0.323	0.371	0.400	0.418	0.431	0.412	0.383	0.338	0.299	0.276	(40)
(41)		taud			2.413	2.370	2.403	2.302	2.287	2.285	2.225	2.285	2.325	2.447	2.489	2.438	(41)
(42)		Ebn at Noon			872	910	923	896	875	857	840	846	845	849	841	841	(42)
(43)		Edh at Noon			77	95	104	124	130	131	138	126	112	88	71	69	(43)
(44)	All-Sky Solar Radiation	RadAvg			1.86	2.88	3.80	4.71	5.58	6.19	6.32	5.43	4.47	2.96	2.00	1.49	(44)
(45)		RadStd			0.20	0.23	0.38	0.42	0.41	0.52	0.37	0.36	0.31	0.38	0.16	0.12	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (53 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page